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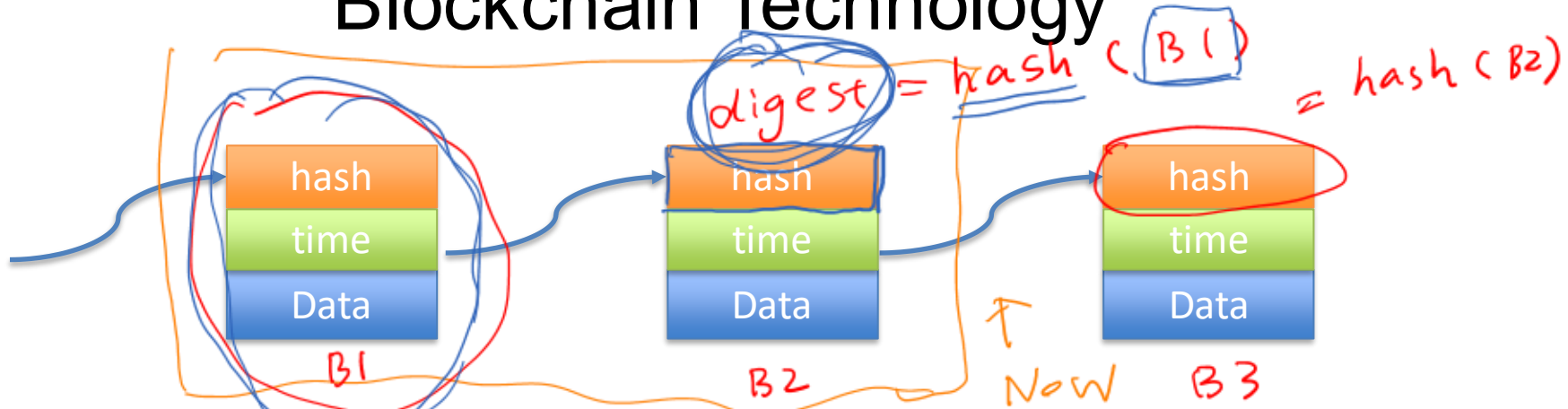
SC6109 – Blockchain Privacy & Scalability

Lecture 01 – Introduction to Privacy Issues

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Blockchain Technology



- A blockchain is a **distributed ledger** with growing list of records that are securely linked together via **cryptographic hashes**.
- Each block contains a **cryptographic hash** of the previous block, a timestamp, and transaction data
- Blockchains are typically managed by a **peer-to-peer** (P2P) network, where nodes collectively adhere to a **consensus protocol**
- Bitcoin (and other cryptocurrencies) is an application of blockchain technology

A normal CPU $\sim 2^90$ in a day

Cryptography Involved

- Hash Function is to ensure blocks are not modifiable
second-preimage resistance. (2^n) , e.g. SHA-256 (2^{256})
given a block data, it is computationally difficult to find data', s.t. $H(\text{data}) = H(\text{data}')$
- Digital signature
Integrity $H: \{0,1\}^* \rightarrow \{0,1\}^n$
Authentication: { id to link between token & owner
actions from owner
- Group signature
authenticate actions and a group, where the id of individual is kept secret to public
"Manager" is able to trace.
Manager doesn't exist in ring signature.

Hash Function

SHA-1 → SHA-2 → SHA-3
broken in 2005 2007 → 2015

Wang Xiaoyun

"n"

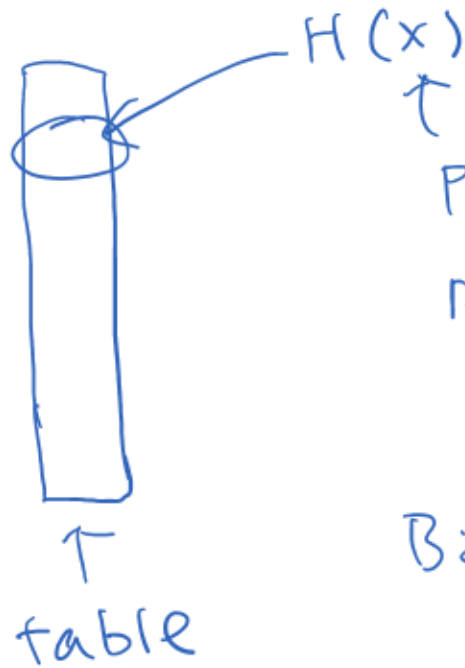
SHA-2: {
SHA-224 } - the first 224 bits of the 256 bits
SHA-256 } output of SHA-256
SHA-384: the first 384 bits of SHA-512
SHA-512

The basic security requirements. $H: \{0,1\}^* \rightarrow \{0,1\}^n$

Collision Resistance: It is computationally hard to find, $x, x' (x \neq x')$, s.t. $H(x) = H(x')$ $(2^{n/2})$

preimage Resistance: Given a ~~random~~ digest d , it is computationally hard to find x s.t. $H(x) = d$ (2^n)

second-preimage Resistance: Given x , it is computationally hard to find $x' (x \neq x')$, s.t. $H(x) = H(x')$ (2^n)



pick randomly

repeat, until a collision is found.

Birthday Attack:

complexity $O(\sqrt{N})$ for

a given space N .

3.10	5.28
4.15	7.12.
2.5	5.26
12.21	8.16
10.20	5.14
7.21	11.16
10.15	9.28
5.4	10.19
10.7	1.10
11.24	4.15
3.19	
11.30	
1.2	

N

birthday bound: $1.4\sqrt{N}$

1. 11.15

2. 9.8

3. 11.24

4. 11.9

5. 5.28

6. 7.10

7. 7.4

8. 2.18

9. 1.9

10. 1.27

11. 8.12

12. 12.30

13. 10.17

14. 7.11

15. 5.12

16. 5.18

17. 11.11

18. 6.24

19. 8.12



RSA and Basics: $\mathbb{Z}_p$ and $\mathbb{Z}_p^*$

$$\begin{aligned}\mathbb{Z}_p &= \{0, 1, 2, \dots, p-1\} \\ &= \mathbb{Z}_p^* \cup \{0\}\end{aligned}$$

$(x, \mathbb{Z}_p^*)$: p is prime

$$\mathbb{Z}_p^* = \{1 \leq x \leq p-1\}, \quad |\mathbb{Z}_p^*| = p-1$$

$$\forall x, y \in \mathbb{Z}_p^*, \quad (\underline{x} \cdot \underline{y} \bmod p) \in \underline{\mathbb{Z}_p^*}$$

$$\text{since } (x, p) = 1, (y, p) = 1, \quad x \cdot y \bmod p \neq 0$$

$$(\mathbb{Z}_p, + \bmod p)$$

Generator g : $\{g^1, g^2, g^3, \dots, g^{p-1}\} = \mathbb{Z}_p^*$

$$\mathbb{Z}_7^* = \{1, 2, 3, 4, 5, 6\}$$

$$g=3, \begin{array}{l} 3^1 = 3 \\ 3^2 = 2 \\ 3^3 = 6 \\ 3^4 = 4 \\ 3^5 = 5 \\ 3^6 = 1 \end{array}$$

3 is a generator of $\mathbb{Z}_7^*$

2 is NOT
generator

$$\underline{2^4} = \underline{2^1} = 2$$

$$g=5, \begin{array}{l} 5^1 = 5 \\ 5^2 = 4 \\ 5^3 = 6 \\ 5^4 = 2 \\ 5^5 = 3 \\ 5^6 = 1 \end{array}$$

5 is also a generator



RSA: the hardness of Factoring Problem

Given N , ^{$(N = p \cdot q)$} it is HARD to find p & q .

$$\sim O(N^{\frac{1}{3}})$$

RSA setup

1. ^{randomly} pick two prime numbers p, q

2. Cal $N = p \cdot q$

$$\varphi(N) = (p-1) \cdot (q-1) = |\mathbb{Z}_N^*|$$

3. randomly pick e , s.t. $(e, \varphi(N)) = 1$

4. Cal $d = e^{-1} \bmod \varphi(N)$ ($e \cdot d = 1 \bmod \varphi(N)$)

(e, N) public key

(d, N) private key

1. $p = 5, \quad q = 7$

2. $N = 35, \quad \varphi(N) = (5-1) \cdot (7-1) = \underline{24}$

3. pick $e = 7$

4. $d = e^{-1} \bmod \varphi(N) \Leftrightarrow d \cdot 7 = 1 \bmod 24$
 $\Downarrow$
 $d = 7$

- https://en.wikipedia.org/wiki/Extended_Euclidean_algorithm

RSA Encryption / Decryption

$$m \rightarrow C$$

$$m^e \bmod N = \underline{\underline{C}}$$

$$C \rightarrow m$$

$$C^d \bmod N = m$$

RSA Signature

Generation

$m, \underline{\text{sign}}$

$$s = \text{sign}(m) = \left(H(m) \right)^d \bmod N$$

(m, H, s) is public

owner (d, e, N)

(e, N)

verification

$$\left(s \right)^e \bmod N \\ \stackrel{?}{=} H(m)$$

1. signature is from the owner of d
2. received (m, H, s) are NOT corrupted.

Q: why the need of H

$$s = m^d$$

$$s^e = m ?$$

If $s_1 = m_1^d$, $s_2 = m_2^d$, then $(s_1 \cdot s_2, m_1 \cdot m_2)$ is valid.

$$(s_1 \cdot s_2)^e = (m_1^d \cdot m_2^d)^e = ((m_1 \cdot m_2)^d)^e = \underline{m_1 \cdot m_2}$$

Forgery Attack, NOT Allowed!

$\mathbb{Z}_p^*$, if $g \in \mathbb{Z}_p^*$, s.t. $\{g^0, g^1, g^2, \dots\} = \mathbb{Z}_p^*$

then g is a generator of $\mathbb{Z}_p^*$.

$$\mathbb{Z}_5^* = \{1, 2, 3, 4\}, g = 2 \text{ or } 3$$

is 1 a generator? $\{1^1, 1^2, 1^3, \dots\} = \{1\} \neq \mathbb{Z}_5^* \times$

2 $\{2^1=2, 2^2=4, 2^3=3, 2^4=1\} = \mathbb{Z}_5^* \checkmark$

3 $\{3^1=3, 3^2=4, 3^3=2, 3^4=1\} = \mathbb{Z}_5^* \checkmark$

4 $\{4^1=4, 4^2=1, 4^3=4, \dots\} \neq \mathbb{Z}_5^* \times$